Supporting information for Patel, Wilcox, Maloney, Ehinger, Patel, & Murray (2025)

1 Figures

Here we describe how to reproduce figures similar to those in the main text.

First run the MATLAB script 'code/a - fits/runme.m'. This script loads the text file 'code/data/data_orientation.csv', which contains trial-by-trial behavioural data from the orientation matching task. It performs the linear regression analysis described in the main text, and saves the results in 'code/data/oslope.csv'. The script also loads 'code/data/data_lightness.csv', which contains trial-by-trial behavioural data from the lightness matching task. It fits the equivalent illuminant model (EIM) described in the main text to this data, and saves the results in 'code/data/eimfit.csv'.

Figures 4 to 12 in the main text show the results of analyses of behavioural data and model observer simulations. To create similar figures, run the suitably named MATLAB script in the Supporting Information, e.g., 'figure4.m'. These scripts rely on the results of the analyses described in the previous paragraph.

We used MATLAB Version 9.14 (R2023a), with the Image Processing Toolbox, the Statistics and Machine Learning Toolbox, and the Optimization Toolbox.

2 Lambertian model of reference patches

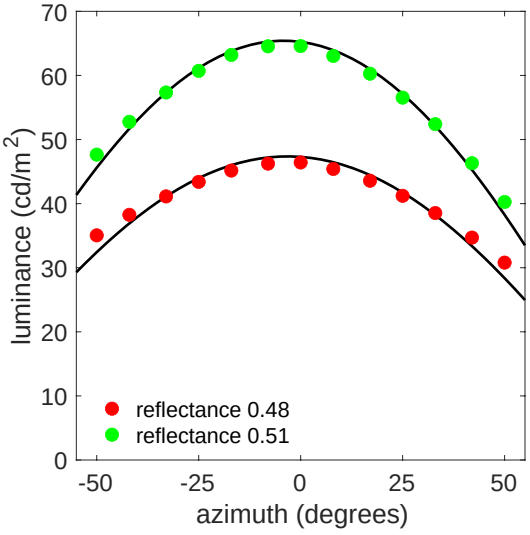


Figure S1: Luminance of the reference patches as a function of azimuth, under frontal collimated lighting (elevation = 0°) in an otherwise dark room. The black lines indicate sum-of-squares fits of $f(\theta; a) = a\cos(\theta)$, corresponding to a Lambertian surface. The reference patches are well-approximated by the Lambertian model.